

Automatic DC-Offset Calibration for a Low-Cost Geophone Front End

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Abstract—This paper presents a digitally assisted foreground calibration method for a low-cost geophone analog front end implemented in a PSoC 5LP mixed-signal system-on-chip. Cascaded gain stages in the front end multiply the offset and drift of the on-chip operational amplifiers, reducing the available dynamic range and eventually driving intermediate nodes into saturation; a fixed correction at the summing node is functional but cannot restore the operating points of the remaining stages. The proposed method sequentially observes four stages – a programmable-gain amplifier (PGA), a band-pass filter (BP), a summing stage (ADDER), and a low-pass filter (LP) – through an on-chip analog multiplexer, reusing the same delta-sigma analog-to-digital converter (ADC) already used for geophone acquisition together with an otherwise-idle digital filter block reconfigured to estimate each stage's DC component. A discrete-time PI controller with deadband and anti-windup adjusts four independent voltage DACs (VDACs) until every stage output approaches its target operating point; neither the controller law nor the DAC realization is intrinsic to the architecture. Once converged, the DAC codes are frozen for acquisition and stored per programmable-gain setting. Oscilloscope measurements on three independently assembled prototype boards, with the PGA at a representative mid-range gain ($\times 16$ of a configurable range up to $\times 50$), show that the BP, ADDER, and LP stages are consistently brought from uncompensated offsets ranging from hundreds of millivolts to about 2 V down to typically a few tens to a few hundred millivolts after calibration; repeated independent runs on two of the boards show a residual run-to-run scatter of comparable order to the correction granularity. The PGA stage instead remains limited by that same granularity rather than by the control loop. A full four-stage run converges in approximately 10 s. The calibrated front end has additionally produced coherent multichannel surface-wave field records, demonstrating operation beyond the bench.

Index Terms—Geophone, analog front end, DC-offset calibration, mixed-signal systems, PSoC

I. INTRODUCTION

Multichannel surface-wave methods use spatially distributed vibration measurements to estimate the dispersive behavior of Rayleigh waves and infer near-surface mechanical properties [1], [2]. Although the processing methods are well established, the acquisition hardware remains a significant part of the cost and deployment complexity, so compact wireless nodes built from inexpensive geophones and mixed-signal microcontrollers are attractive for dense, reconfigurable arrays [3].

The present work forms part of a low-cost Internet-of-Things acquisition platform based on SM-24 moving-coil geophones [9]. A PSoC 5LP was selected as the analog front end because it integrates programmable-gain amplifiers, operational amplifiers, analog filters, voltage DACs, analog multiplexers, a delta-sigma ADC, a digital filter block, DMA, and an Arm Cortex-M3 processor in one device [4]. The guiding principle is to reuse this already-available hardware rather than add dedicated calibration components: the same ADC and digital filter are time-shared between acquisition and calibration, so the correction loop uses only resources the platform already carries, avoiding a separate converter or processor. The same reconfigurability also imposes a limitation: its internal analog blocks cannot be independently optimized for low offset, drift, and flicker noise – Lopin and Lopin reported the same tradeoff in a single-chip PSoC potentiostat, with poorer offset and noise specifications than purpose-selected external amplifiers [5].

The geophone conditioning circuit is derived from the cascade correction and subtraction topology of Ma *et al.* [6]; mapped to the PSoC, accumulated DC errors prevented sufficiently aggressive gain settings and became particularly severe at the ADDER output.

A continuously active DC servo, evaluated as an initial solution, exposed an unavoidable compromise: a low bandwidth centered impractically slowly, a faster loop disturbed the signal band, and every gain change forced a new transient. The proposed method instead performs a finite foreground calibration before measurement and disconnects the controller afterward.

Self-calibration using on-chip resources is not itself new. The PSoC-Stat injects five known current levels with an IDAC and uses the resulting ADC values to estimate the gain and offset of a TIA/ADC signal chain [5]. Infineon's AN68403 applies an internal VDAC to a PGA/ADC chain, measures the system offset with grounded inputs, and stores gain and offset corrections in EEPROM [7]. These methods primarily correct the numerical transfer characteristic observed at the ADC output. The objective here is different: the proposed loop changes the physical reference of each amplification stage so that its output is adjusted toward its intended operating point, within the available DAC range and resolution.

The available ADDER made it possible to implement a fixed DC injection as an early workaround, viable for this circuit but not generally transferable to front ends without an explicit summing node. More importantly, it is only a patch: earlier stages keep drifting before the correction point, while later stages introduce new offsets after it.

The contribution is threefold: 1) a per-stage analog calibration architecture that reuses the PSoC analog multiplexer, ADC, digital filter, and four VDACS; 2) a controller-independent, measurement-gated foreground implementation, demonstrated with a gain-aware PI controller using deadband, anti-windup, slew limiting, stability detection, and one-LSB refinement, that freezes all corrections before seismic data are recorded and so introduces no additional closed-loop poles into the acquisition transfer function; and 3) an experimental validation on three independently assembled boards at maximum PGA gain, showing consistent offset reduction on the BP, ADDER, and LP stages and identifying correction granularity as the limiting factor on the PGA stage.

II. PLATFORM DESIGN

A. Processing Architecture

The platform was selected for low cost, reconfigurable analog routing, hardware-assisted signal processing, reuse of acquisition resources across operating modes, and sufficient conversion resolution for both measurement and internal diagnostics. In acquisition mode, the PSoC conditions and digitizes the geophone output; in calibration mode the input chain is unchanged, but the observation path, digital-filter coefficients, and DAC-control firmware reconfigure to close a temporary mixed-signal loop.

For the i th stage, its slowly varying output can be represented as

$$y_i^{\text{DC}} = A_i y_{i-1}^{\text{DC}} + b_i + K_i u_i, \quad (1)$$

where A_i is the stage gain, b_i the amplifier and reference errors, K_i the signed transfer from the trim DAC to the observed node, and u_i the DAC code. Adjusting only the ADDER reference forces one output toward its target but does not constrain the remaining y_i^{DC} values; an earlier rail cannot be recovered this way, and offsets appearing after the ADDER stay uncompensated.

The proposed architecture instead chooses each u_i in Eq. 1 independently so that every observed y_i^{DC} approaches its stage reference. In the differential geophone front end considered here (GEO), all four calibration targets correspond to zero ADC counts.

III. SELF-CALIBRATING GEOPHONE PLATFORM

A. Signal Conditioning and Observation Circuit

The final GEO prototype, shown in Fig. 1, contains four calibrated points: the output of the programmable-gain amplifier (PGA), the band-pass filter (BP), the summing stage (ADDER), and the low-pass filter (LP), each with an independent 8-bit VDAC. An analog multiplexer connects one point at a time to the same 18-bit delta-sigma ADC used to digitize the geophone

signal; its readings are rescaled into VDAC-code-equivalent units, letting the acquisition converter serve as the error sensor without a second ADC; software-enforced mutual exclusion prevents two analog outputs from being connected at once.

The ADC observation path includes a switched capacitor behind the selected analog-multiplexer MOSFET, whose on-resistance forms a limited RC anti-alias network while isolating the amplifier output from a direct capacitive load. The PSoC digital filter block is also reused: during acquisition it implements the geophone filter, and during calibration its coefficients switch to an aggressive low-pass response that provides a stable DC-level estimate. Hardware reuse here reduces CPU intervention and preserves the existing acquisition architecture; acquisition coefficients and signal path are restored at the end of calibration.

B. Digitally Assisted Control Calibration

For each stage, the filtered observation $\hat{y}_i[n]$ produces the ADC-domain error $e_i[n] = r_i - \hat{y}_i[n]$, where $r_i = 0$ for the current differential GEO configuration. The error is first mapped to the 8-bit DAC domain:

$$\varepsilon_i[n] = \text{round}\left(\frac{e_i[n] V_{\text{ADC}} N_{\text{DAC}}}{N_{\text{ADC}} V_{\text{DAC}}}\right), \quad (2)$$

with $V_{\text{ADC}} = 5000$ mV, $N_{\text{ADC}} = 262144$, $V_{\text{DAC}} = 4080$ mV, and $N_{\text{DAC}} = 255$ in the implemented configuration.

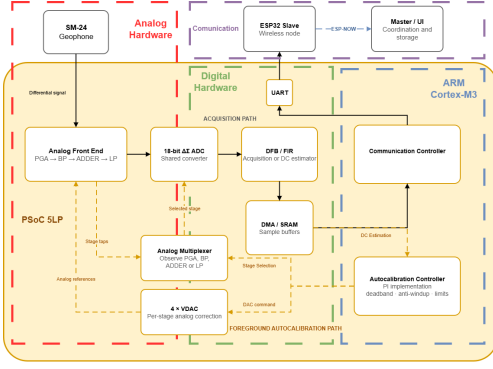
The calibration architecture only requires the selected stage be observable and the trim DAC provide sufficient control range, so the controller can be chosen per application. In this prototype, a positional PI law was used for its simplicity:

$$u_i^*[n] = u_{i,0} + s_i \left[\frac{k_{p,i}}{|K_i|} \varepsilon_i[n] + \frac{k_{i,i}}{|K_i|} \sum_{q=0}^n \varepsilon_i[q] \right], \quad (3)$$

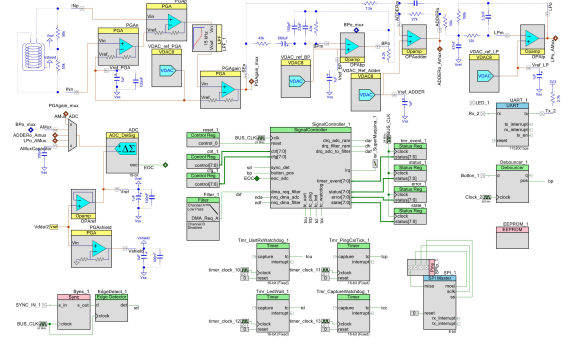
where $u_{i,0}$ is the feedforward code, an initial DAC value near stage i 's expected operating point; $s_i \in \{-1, +1\}$ is the injection polarity; and $K_i > 0$ is the magnitude of the DC gain from the DAC's injection point to the shared observation node – i.e., how much a static VDAC code change at stage i moves the observed node, expressed in ADC-equivalent counts per VDAC code (Eq. 2). K_i is a nominal design value, fixed for BP, ADDER, and LP (2, 7.89, and 6, reflecting downstream gain, not individually measured per board); for PGA, K_i and s_i are recomputed from the active gain code.

The command $u_i^*[n]$ of Eq. 3 is saturated to the valid VDAC interval and limited to one code per sample, with conditional integration for clamping anti-windup. The deadband is $\Delta_i = \max(1, \lceil 1.2|K_i| \rceil)$ DAC codes; convergence requires the error to stay in the same quantized bucket for a configurable number of samples, and a final adjacent-code trial is kept only if it reduces the filtered error. If a stage misses its deadband before a per-stage timeout, the DAC stays at the best code found, the stage is flagged not converged, and the sequence moves on, bounding worst-case time without leaving any stage arbitrary or saturated.

Table I lists the parameters used, to support reproduction. Settle is 512 samples (0.20 s) for PGA/BP/ADDER and 1024



(a) System block diagram.



(b) PSoC Creator TopDesign implementation.

Fig. 1: Proposed analog front end and digitally assisted calibration path: (a) system-level view of the acquisition and foreground calibration paths; (b) the corresponding hardware implementation in the PSoC 5LP, including the analog front end, the shared ADC/multiplexer/VDAC observation-and-correction path, and the digital blocks (DMA, hardware filter, timers) that time-share acquisition and calibration.

TABLE I: Implemented per-stage calibration parameters. The DC estimator runs at the ADC native sample rate of 2604 Hz.

Stage	K_i	$k_{p,i}$	$k_{i,i}$	Timeout (s)
PGA	gain-dependent	10^{-3}	3×10^{-4}	11.5
BP	2	10^{-4}	5×10^{-4}	17.3
ADDER	7.89	10^{-4}	5×10^{-4}	11.5
LP	6	10^{-4}	5×10^{-4}	17.3

(0.39 s) for LP; lock requires 512 or 1024 consecutive in-deadband samples, and refinement settles a further 1024–2048 samples. Timeouts are worst-case ceilings, not the typical duration shown in Fig. 3.

The 8-bit VDAC resolution suffices here; a finer step could use a PSoC IDAC/resistor or a buffered topology [8] – an implementation choice, not a limitation, since controller, range, and step size can all adapt while keeping the same observe–estimate–correct sequence.

C. Embedded Sequence and Persistence

At boot, after a gain change, or upon command, the firmware loads the stored DAC values, verifies the resulting offsets, and restores the measurement path without a full run if every stage is within tolerance. Otherwise it processes PGA, BP, ADDER, and LP sequentially, settling after each multiplexer switch and FIR-history reset; a watchdog and per-stage sample limit bound the run.

Successful codes are stored in EEPROM, one record per PGA setting (hardware class, gain code, stage/valid-stage mask, four DAC codes, version, CRC); an unchanged setup only needs to load and verify a prior record, and a failed run never overwrites a valid one.

During acquisition the PI is inactive, the observation multiplexer is returned to the LP output, and all VDAC codes remain constant. Thus, the method changes the DC operating point but does not introduce a time-varying feedback loop into the acquired signal. Fig. 2 summarizes the complete boot-to-acquisition sequence described in this section.

IV. EXPERIMENTAL VALIDATION

A. Prototype and Comparison Conditions

The validation compares two configurations on three independently assembled SM-24 GEO prototypes (Board A, Board B, and Board C), each containing the four independently controlled VDAC references: *Default*, fixed nominal VDAC references without calibration, and *Proposed*, sequential four-stage calibration followed by frozen-reference acquisition.

The earlier ADDER-only workaround is not a separate measured condition: it only recenters the ADDER node, equivalent to calibrating that stage in isolation, and would not expose drift before/after it, precisely what this comparison isolates. All measurements used the PGA at $\times 16$, a mid-range setting chosen to bound the experimental scope; the PGA is configurable up to $\times 50$, and validating higher, more demanding gains is future work.

B. Measured Offset Reduction Across Three Prototypes

For each board, the DC offset at the output of the PGA, BP, ADDER, and LP stages was measured with an oscilloscope, both with the default (uncalibrated) VDAC references and after the proposed calibration converged. Default and Proposed are reported as mean $\pm$ SD across repeated independent runs, $N=5$ on Boards A and B; Board C has only $N=1$ so far, shown in red in Table II pending further runs. Sign differences reflect each stage’s injection polarity (Eq. 3): PGA is approached from below zero, BP/ADDER/LP from above.

On all three boards, BP, ADDER, and LP show a large offset reduction after calibration, consistent with the order-of-volts uncompensated drift seen during earlier bench work; the ADC’s ± 2.5 V range (Eq. 2) leaves headroom to observe the largest uncalibrated offset in Table II (−1.99 V, Board A LP minimum) without saturating. The PGA stage differs: its post-calibration mean is not consistently closer to zero than Default, and its run-to-run standard deviation is comparable to or larger than the other stages – a correction-granularity

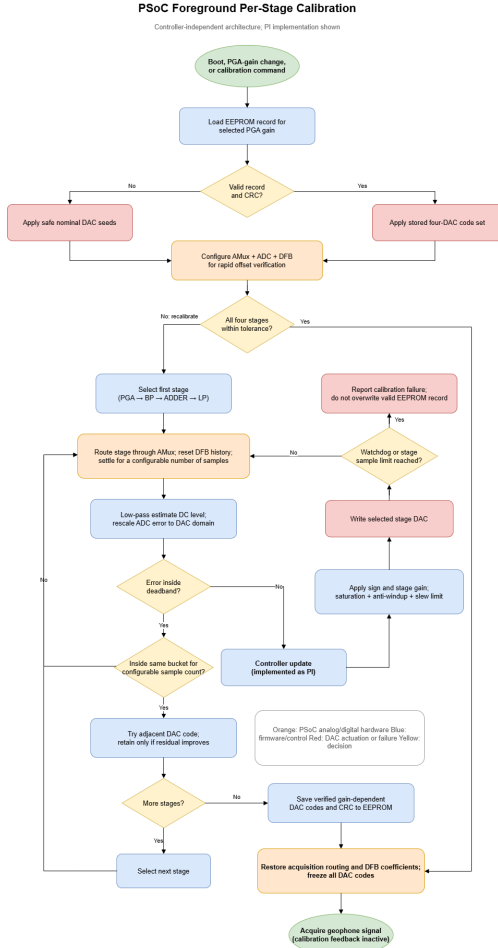


Fig. 2: Flowchart of the proposed foreground autocalibration algorithm, including stored-code verification, sequential per-stage correction, and transition to acquisition.

limitation, since a residual under roughly one PGA step is invisible inside the deadband of Eq. 2; a finer element such as [8] could tighten this. The repeated runs on Boards A and B show the BP, ADDER, and LP residuals, while always far smaller than the uncalibrated offset, vary between runs: the standard deviations (17–87 mV) are of the same order as the deadband Δ_i (Section III), consistent with quantization rather than drift. Default readings also varied between runs on the same board – e.g., Board B PGA ranged from -1.7 mV to over 200 mV, and Board A ADDER changed sign across runs – indicating the uncalibrated baseline is not stable enough for a single factory trim, part of the motivation for calibrating on every boot (Section V). Confirming these trends with $N \geq 20$ runs per board on a larger population is left as future work.

C. Calibration Transient

Fig. 3 shows an oscilloscope capture of the four observed stage outputs during an actual run, spanning the sequential $\text{PGA} \rightarrow \text{BP} \rightarrow \text{ADDER} \rightarrow \text{LP}$ correction. Each output steps as the controller reaches it and holds once converged, visible as the successive plateaus in the ADDER and LP traces; the full sequence completes in approximately 10 s.

TABLE II: Per-stage DC offset before (Default, mean $\pm$ SD [min, max]) and after (Proposed, mean $\pm$ SD) calibration, PGA at $\times 16$. Board C (red) has only one run; repeatability there is pending.

Board	Stage	Default (mV)	Proposed (mV)
A	PGA	110.8 ± 136.8 $[-126, 203]$	106.9 ± 63.0
	BP	302.4 ± 79.1 $[221, 411]$	107.1 ± 87.0
	ADDER	232.1 ± 334.8 $[-159, 594]$	-6.4 ± 54.3
	LP	-1360.6 ± 566.5 $[-1990, -606]$	133.8 ± 52.5
B	PGA	151.3 ± 86.5 $[-1.7, 204]$	132.8 ± 101.8
	BP	501.6 ± 13.0 $[480, 512]$	87.6 ± 16.7
	ADDER	270.4 ± 46.8 $[228, 338]$	-87.0 ± 63.4
	LP	737.8 ± 98.8 $[623, 884]$	289.0 ± 76.1
C	PGA	202 (falta medir)	206 (falta medir)
	BP	374 (falta medir)	213 (falta medir)
	ADDER	897 (falta medir)	48.8 (falta medir)
	LP	342 (falta medir)	31.9 (falta medir)

Runs/board: A $N=5$, B $N=5$, C $N=1$ (pending).

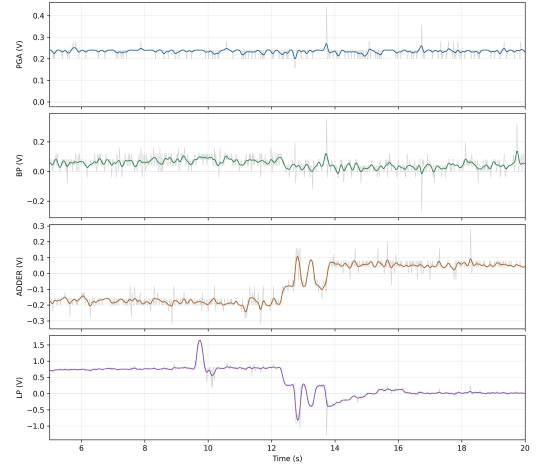


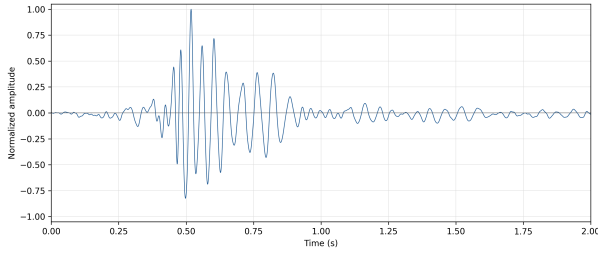
Fig. 3: Oscilloscope capture of the PGA, BP, ADDER, and LP stage outputs during a single calibration run at PGA gain $\times 16$. Uncalibrated until $t \approx 9$ s, all four stages converge by $t \approx 16$ s. PGA shows a visibly larger residual, consistent with its coarser correction granularity (Section IV). Traces are individually labeled and remain distinguishable in grayscale.

D. Field Use of the Calibrated Front End

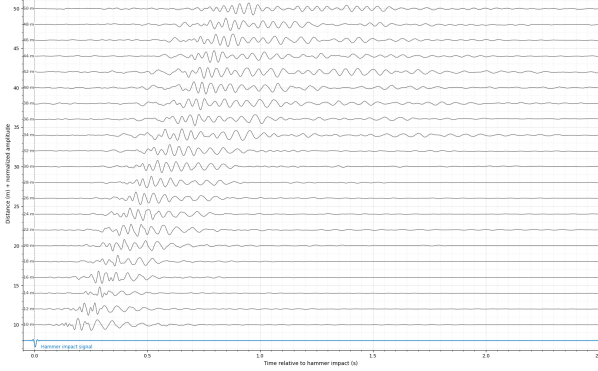
The calibrated front end has also been exercised in field surface-wave surveys. Fig. 4 shows (a) a normalized SM-24 hammer response through the calibrated channel, and (b) a multichannel waterfall across a 10–50 m line, with first-arrival moveout resolved at every offset using the higher PGA settings needed far-offset. These are field evidence of correct operation, not a controlled calibrated-vs-uncalibrated comparison, which remains future work.

V. DISCUSSION

The method trades component precision for system-level observability and digital assistance. Its benefit is not that the ADC output has zero mean – a software subtraction could do that – but that the DACs modify the physical operating points before large downstream gains apply, so the correction



(a) Single-channel hammer response.



(b) Multichannel waterfall, 10–50 m.

Fig. 4: Field records acquired with the calibrated front end.

is agnostic to the error source, provided the stage is observable and the DAC has sufficient authority and range.

The foreground method was adopted after a continuously active DC servo showed a low corner centered impractically slowly, a faster loop interfered with the signal, and a gain change forced another transient. The proposed controller instead disconnects, holding DAC outputs constant throughout acquisition, adding neither time-varying feedback nor extra servo poles; EEPROM records avoid the full transient after most gain changes. Since the DC estimator is an aggressive low-pass filter, transient seismic energy is itself attenuated, so the method should tolerate a non-silent site given no sustained disturbance during convergence, though this was not validated under controlled vibration. The method corrects the offset present when calibration runs, not drift directly; recalibrating at boot, gain change, or on command (Section III) limits drift accumulation, though temperature/time-resolved drift was not characterized. Remaining costs are finite DAC resolution and an inability to correct noise, distortion, or an exhausted range.

The PI implementation completes a full four-stage run in approximately 10 s (Fig. 3); a faster controller could replace it without changing the observation and correction hardware, but a simple PI was preferred since calibration triggers once after deployment, so a one-time delay of this order is acceptable.

Three boards of the final four-VDAC revision were available for this work; measuring both conditions on all three shows the offset reduction of Table II is not particular to one assembly, while unit-to-unit PGA differences (Board C vs. Board B) interact with the correction granularity above. Repeated runs on Boards A and B (Section IV) begin to separate algorithmic repeatability from board-to-board variation, but a larger popu-

lation, matching repetition on Board C, and cross-temperature trials remain necessary before generalizing it. As noted in Section I, the comparison with prior self-calibration work [5], [7] is architectural, not quantitative, since those methods target the ADC transfer characteristic rather than each stage’s physical operating point.

VI. CONCLUSION

A per-stage foreground self-calibration method was developed for a low-cost PSoC-based geophone front end. Four internal VDACS, an analog multiplexer, the acquisition delta-sigma ADC, an otherwise-idle hardware filter, and a configurable PI controller adjust each observable stage’s operating point within the available DAC range and resolution. Unlike an ADDER-only patch, the method addresses offsets before and after the summing node; unlike the continuously active servo tested during development, it adds no servo poles or settling-time-versus-distortion compromise. Stored gain-dependent corrections make full recalibration infrequent, while automatic verification restores the measurement path without a full run when conditions have not changed.

Oscilloscope measurements on three boards, PGA at $\times 16$ (mid-range; configurable to $\times 50$), showed the BP, ADDER, and LP stages consistently brought from an uncompensated offset of hundreds of millivolts to a few volts down to a few tens to a few hundred millivolts; repeated runs on two boards show this is repeatable, with run-to-run scatter on the order of the correction granularity, which also limits the PGA stage rather than the control loop. Calibration converges in approximately 10 s per run, acceptable since triggered once after deployment; field surface-wave records over a 10–50 m line demonstrate operation, not a quantitative comparison. A larger board population, remaining gain settings, intermediate-node headroom, drift behavior, and a controlled calibrated-vs-uncalibrated comparison are future work.

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